Name & Surname **Polimi ID**

Exercise 1

A. On a PRAM model, the time complexity can be equal to the number of instructions executed. True/False? Why? (1)

B. According to the definitions we have seen during the PRAM lecture, how could the speedup be put in relation to the efficiency? (1)

C. Please describe how the matrix vector multiplication can be parallelized on a PRAM model. Please report also speedup, efficiency, work and cost. (1)

D. Among the laws we have seen during the course, which one is classified to address the weak scaling? Please also describe the assumptions that come with the identified law. (1)

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Exercise 2

A. Briefly describe what “Divergent” execution” means. Provide also an example to support the description. (1)

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B. Describe the difference between the Message Passing and Shared Address Space programming model. (1)

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C. Can a program with many arithmetic operations and a small number of memory accesses be a problem for a multi-threaded processor? True/False. Why? (1)

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D. Please compare Coarse-grain multithreading with Simultaneous multithreading. (1)

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Exercise 3

A. Describe the distinct types of address spaces visible to kernels in a CUDA/GPU based environment. How many? How much is shared? How fast are they? (2)

B. Briefly describe how CUDA threads-block are assigned to hardware considering the V100 SM processor architecture discussed during the first part of the course. (2)

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Exercise 4

A. What happens when a variable is declared as firstprivate in an OpenMP pragma? (1)

B. What is the difference between a static and a dynamic schedule in the OpenMP for construct? (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

C. Write a small example of nested parallelism in OpenMP and clearly indicate how many threads execute each region. (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**